Addis Ababa University
አዲስ፡አበባ፡ዩኒቨርሲቲ
SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !

COLLEGE OF NATURAL AND COMPUTATIONAL SCIENCES

MASTERS OF SCIENCE IN BIOINFORMATICS

ADVANCED PROGRAMMING FOR BIOINFORMATICS

Assignment 1

Name: Tsion Legesse

ID No: GSR/1433/18

Department:MOLECULAR SCIENCE AND GENETICS

Course Code:BINF6182

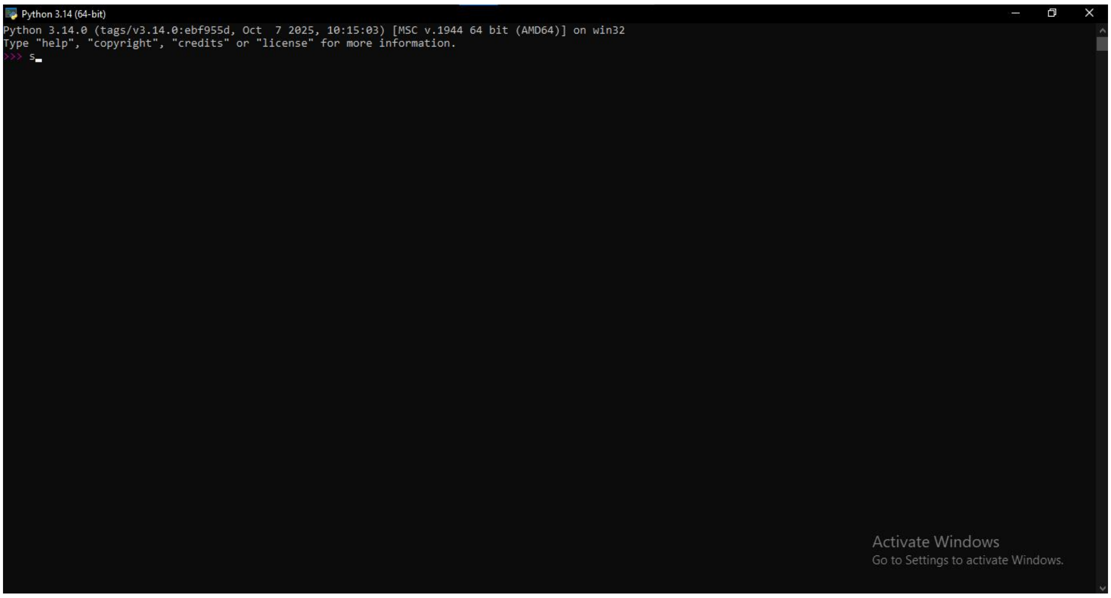
Submission date: April 24 2026
Submited to: Nigatu (PhD)

ANSWER

Task 1: Environment Setup

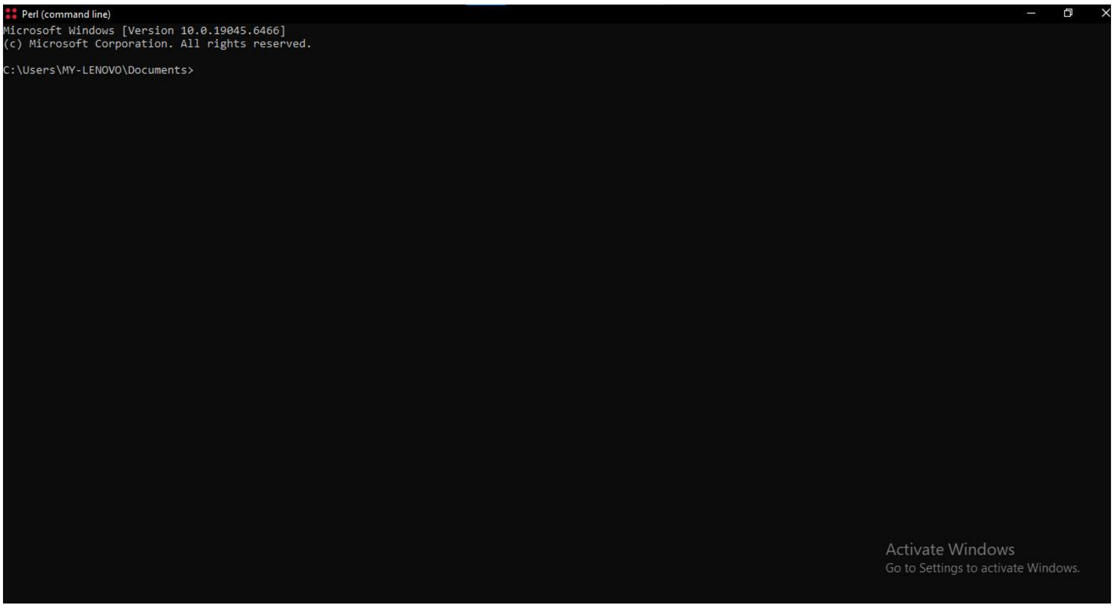
Python (version 3.x)

Python is a high-level, easy-to-learn programming language known for its simple syntax. It is widely used in data analysis, machine learning, web development, and automation. Its large library ecosystem makes it very versatile.



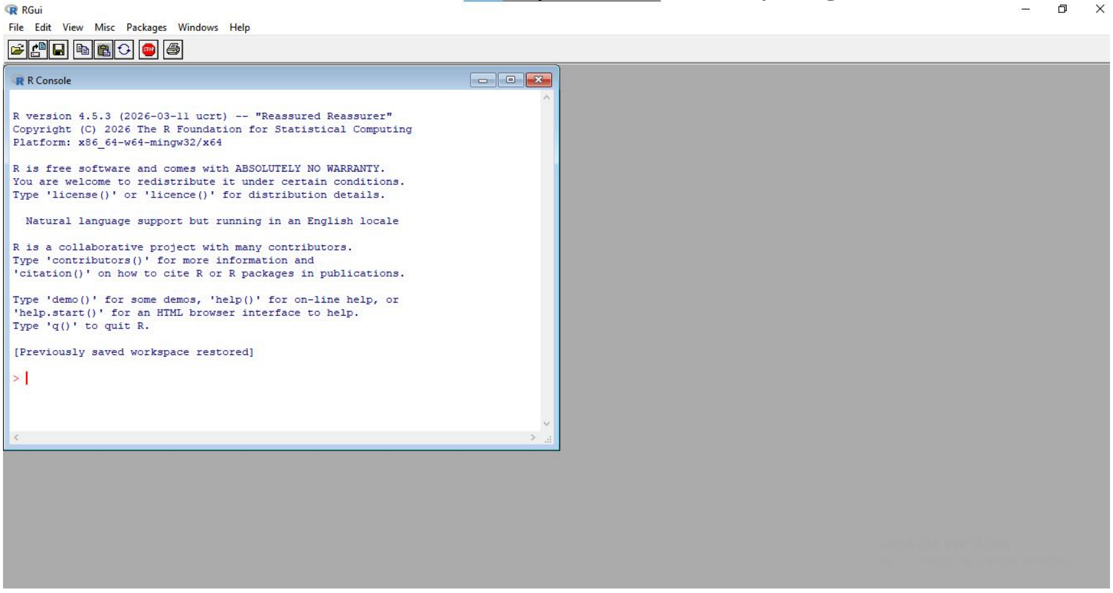
• **Perl**

Perl is a scripting language designed for powerful text processing and pattern matching. It is commonly used in bioinformatics and system administration tasks. Perl is especially good at handling large amounts of text data.



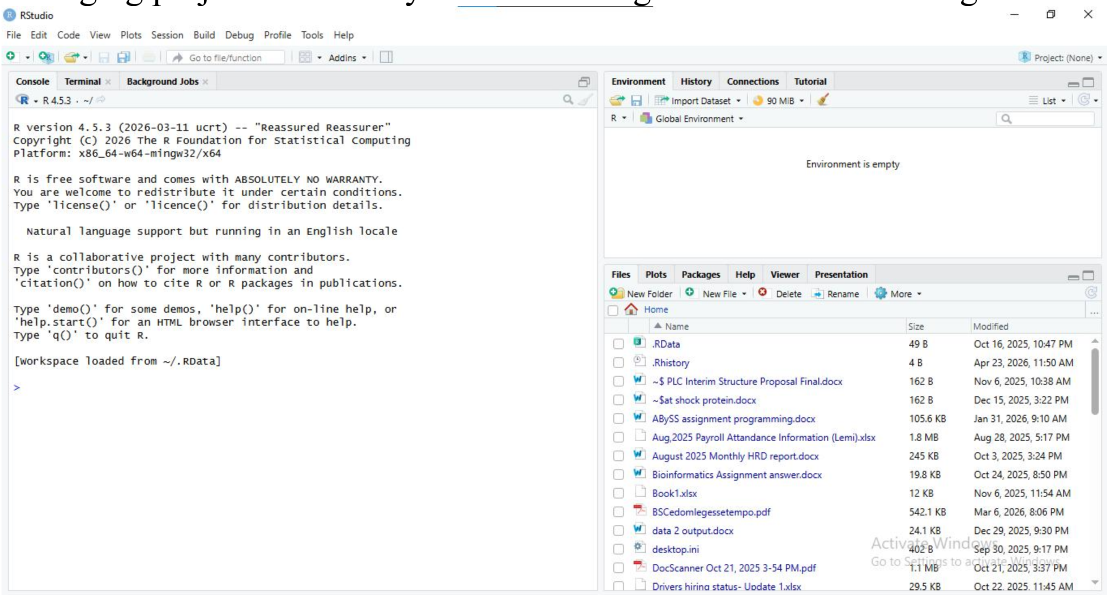
• **R**

R is a programming language specifically built for statistical computing and data analysis. It provides strong support for data visualization and modeling. Researchers and data scientists widely use it for analyzing datasets.



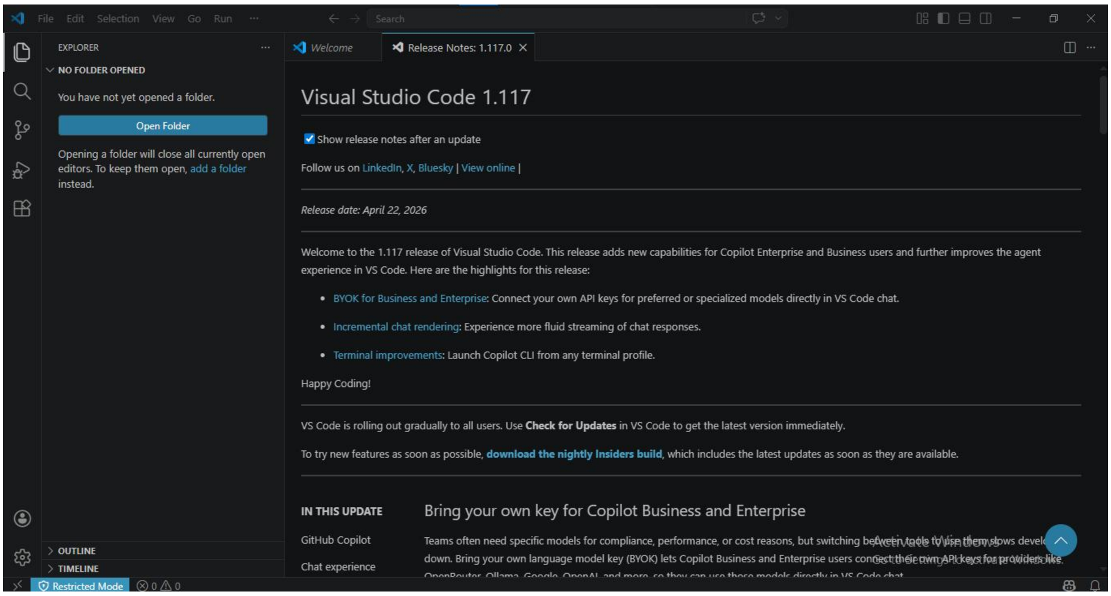
R STUDIIO

RStudio is an integrated development environment (IDE) designed for working with R. It provides tools for writing code, visualizing data, and managing projects efficiently. It makes using R easier and more organized.



• **Text editor (VS Code)**

Visual Studio Code is a lightweight and powerful code editor developed by Microsoft. It supports many programming languages through extensions, including Python and R. It offers features like debugging, syntax highlighting, and version control integration.



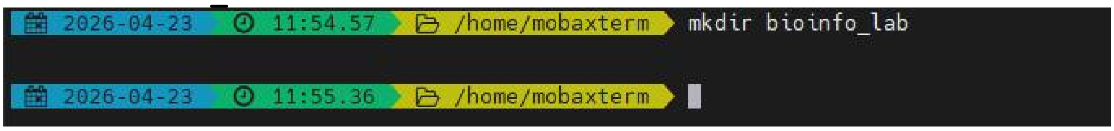
Task 2: Linux File System Navigation

Perform the following:

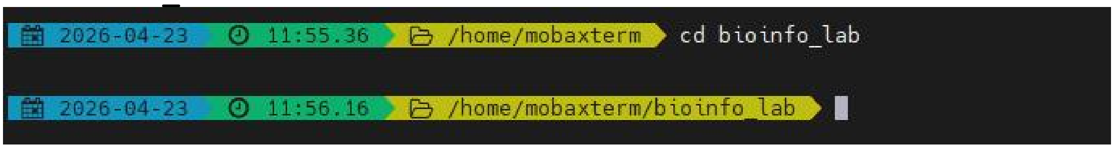
pwd

ls

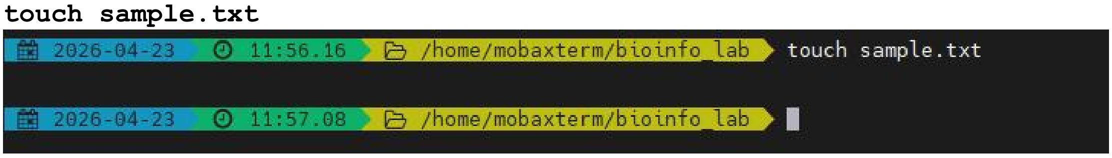
mkdir bioinfo_lab



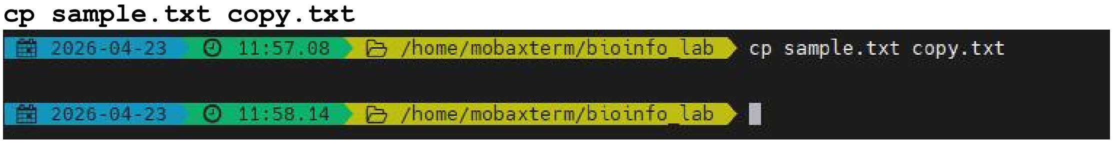
cd bioinfo_lab



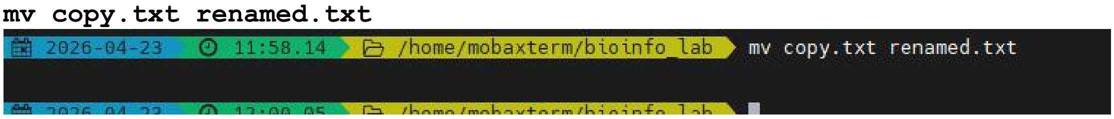
touch sample.txt



cp sample.txt copy.txt



mv copy.txt renamed.txt



What does pwd do?

pwd stands for print working directory. It shows the full path of the directory you are currently in.

Difference between cp and mv?

- **cp (copy)** creates a duplicate of a file or folder, leaving the original unchanged.
- **mv (move)** either moves a file to a new location or renames it, removing it from the original place.

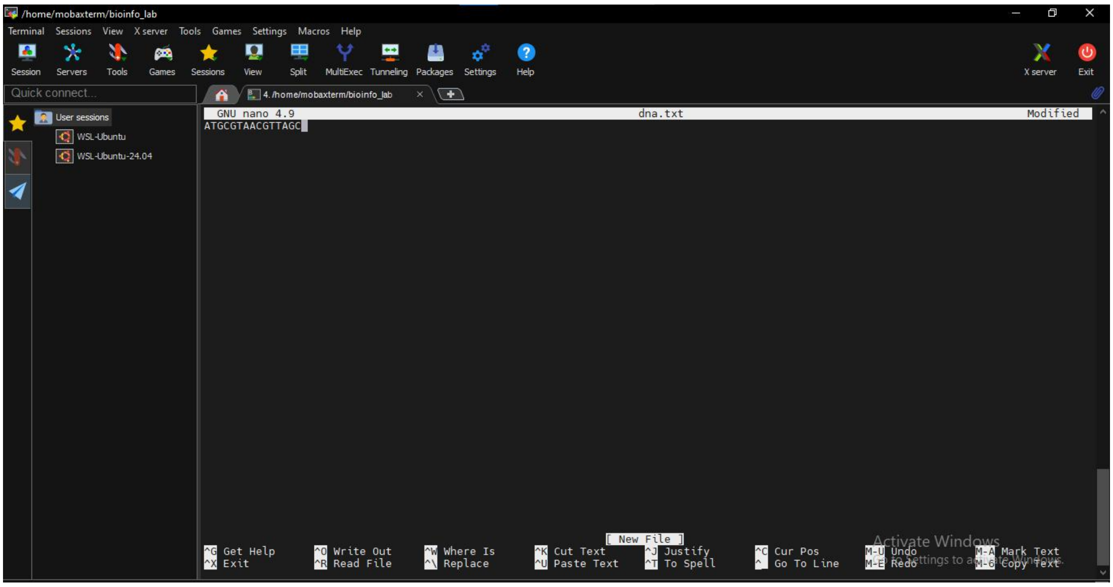
Task 3: File Handling & Text Processing

Create a file:

nano dna.txt

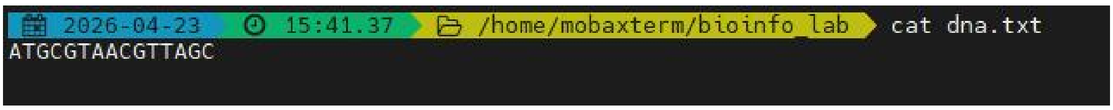
Add:

ATGCGTAACGTTAGC

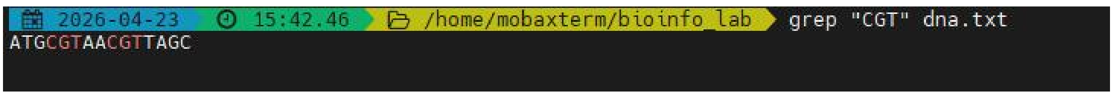


Perform:

cat dna.txt



grep "CGT" dna.tx



What does grep do?

grep is a command used to search for a specific pattern or text inside a file. It prints the lines that contain the matching pattern

The output is:

ATGCGTAACGTTAGC

This is because the sequence contains **"CGT"**, so grep returns the entire line where the match is found.

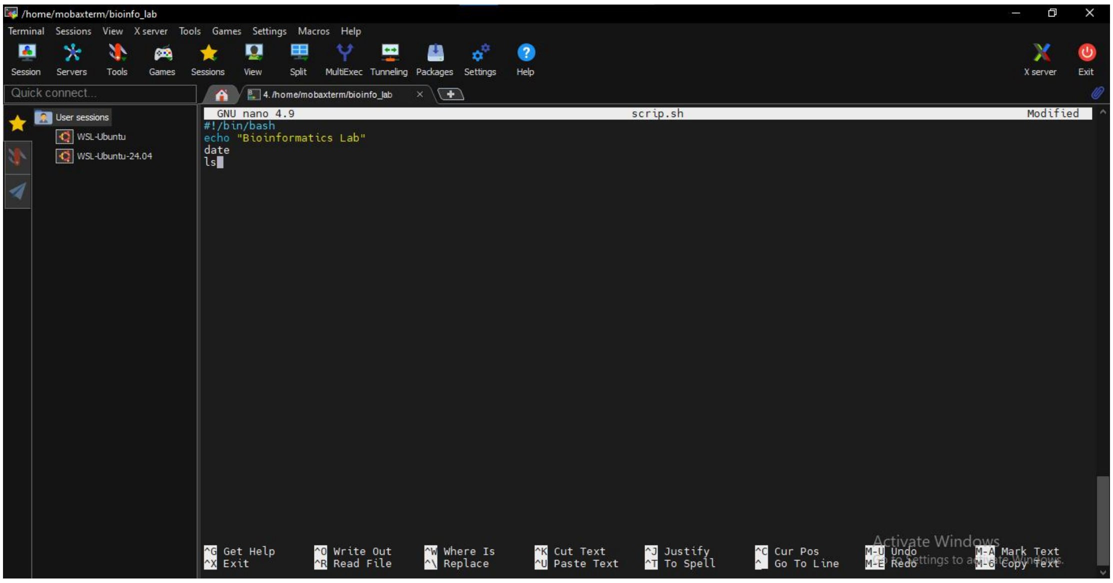
Task 4: Shell Scripting

Create script:

```
nano script.sh
```

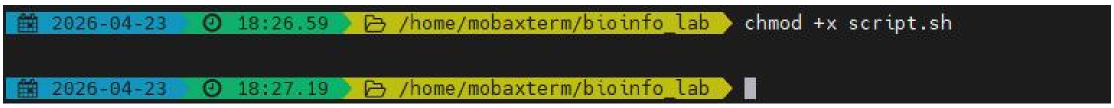
Add:

```
#!/bin/bash
echo "Bioinformatics Lab"
date
ls
```

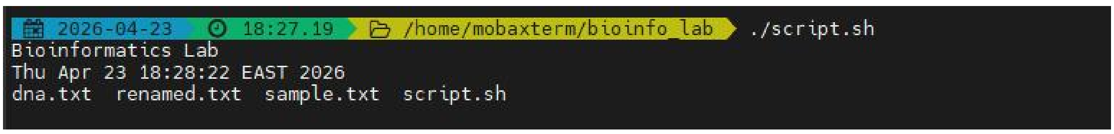


Run:

```
chmod +x script.sh
```




```
./script.sh
```



Explanation of Each Command

#!/bin/bash

This is called a **shebang**. It tells the system to use the Bash shell to run the script.

echo "Bioinformatics Lab"

Prints the text **Bioinformatics Lab** to the terminal.

date

Displays the current system date and time.

ls

Lists all files and directories in the current folder

chmod +x script.sh

Gives the script **execute permission**, so it can be run

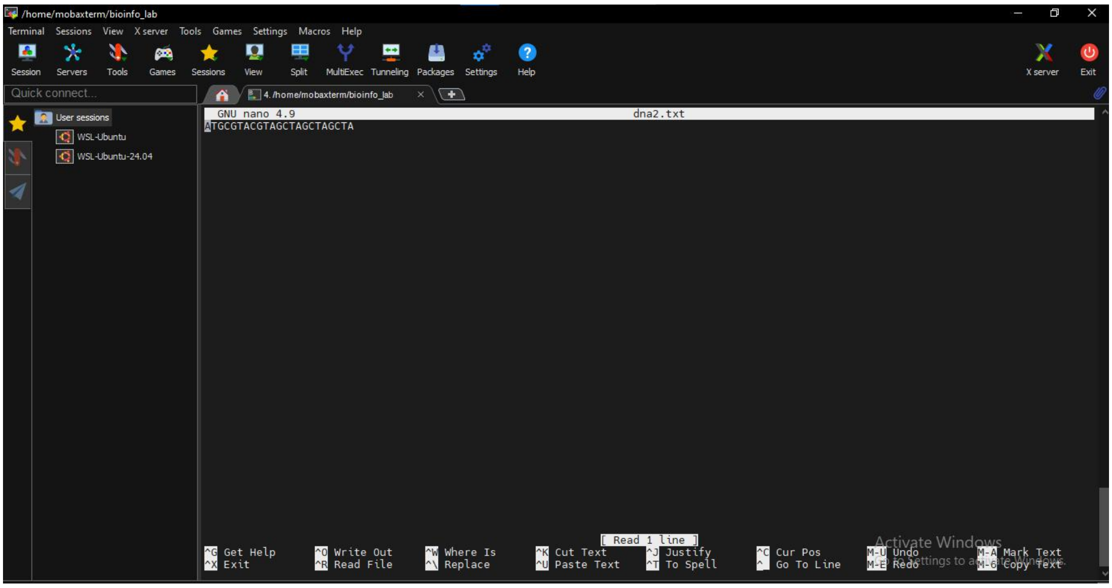
./script.sh

Runs the script from the current directory.

Task 5: Simple Bioinformatics Task

Create a DNA file:

```
ATGCGTACGTAGCTAGCTAGCTA
```



Write a shell command to:

• Count number of characters

```
2026-04-23  21:15.58  /home/mobaxterm/bioinfo_lab  wc -c dna2.txt
24 dna2.txt
```

• Count occurrences of "A"

```
2026-04-24  06:30.33  /home/mobaxterm/bioinfo_lab  grep -o "A" dna2.txt | wc -l
6
```

Interpretation

- The DNA sequence contains **23 actual characters (nucleotides)**.
- The command wc -c returns **24** because it also counts the hidden newline (\n).
- The nucleotide **"A" appears 6 times** in the sequence.
- grep -o "A" isolates each occurrence, and wc -l counts them accurately.